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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination - APRIL2021

AMT202S – ALLIED MATHEMATICS - II

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Prove that $\text{Cosh}^2 x - \text{Sinh}^2 x = 1$.
2. Find the complete integral $p + q = pq$.
3. Define Solenoidal.
4. State Stoke's theorem.
5. Write the Newton's Backward interpolation formula.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Find $\log(1 - i)$.
7. Solve $p(1+q) = qz$.
8. Find the directional derivative of $\phi(x, y, z) = xy + yz + zx$ at the point (1, 1, 3) in the direction of $\vec{i} + 2\vec{j} + 2\vec{k}$.

9. Show that, if S is a closed surface, then $\iint \vec{r}.\hat{n}ds$ =3V, where V is the volume enclosed by S.

10. Use Newton's formula to find of when $x = 142$ given that

x	140	150	160	170	180
y	3.685	4.854	6.302	8.076	10.225

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) Prove that $64 \sin^7 \theta = 35 \sin \theta - 21 \sin 3\theta + 7 \sin 5\theta - \sin 7\theta$.

(or)

b) If $\cos (x + iy) = \cos \theta + i \sin \theta$ the prove that $\cos 2x + \cos h2y = 2$.

12. a) Solve $p^2 + q^2 = x + y$

(or)

b) Solve $(y^2 + z^2)p - xyq = -xz$

13. a) If $F = xyz\,\vec{i} + xyz^2\,\vec{j} + x^2 yz\,\vec{k}$ then find div curl F and curl of curl F.

(or)

b) Find $\nabla.(\nabla r^n)$ and $\nabla \times \nabla r^n$.

14. a) Verify Green's theorem $\int (x - 2y)dx + xdy$, where C is the circle $x^2 + y^2 = 1$

(or)

b) Evaluate $\iint x^2 dydz + y^2 dzdx + 2z(xy - x - y)dxdy$, where S is the surface of the cube $0 \le x \le 1$, $0 \le y \le 1$, $0 \le z \le 1$.

15. a) Estimate the population for the year 1925 from the data given below

Year(x)	1891	1901	1911	1921	1931
Population (y)	46	66	81	93	101

(or)

b) Using Lagrange's interpolation formula find f(x) from the following table.

x	1	2	3	4	7
y = f(x)	2	4	8	16	128

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